Laboratory work No. 1. Linear Regression.

1. **Dataset**

According to the internal code download the data set for which you will need to build a linear regression model to predict a target feature. If the dataset contains more than 20 features, select 20 random features (there should be both categorical and numerical features).

No.	Internal code	Dataset	Target variable
1	F0433	Medical Cost Personal Datasets	charges
2	F4857	Used Car Price Prediction	price_usd
3	F8623	Melbourne Housing Market	Price
4	F8633	Philippines Housing Market (Housing_v2.csv)	Price
5	F0494	Quebec Housing Sales price	sale_price
6	E8117	Medical Cost Personal Datasets	charges
7	F0221	Used Car Price Prediction	price_usd
8	F8794	Melbourne Housing Market	Price
9	F8810	Philippines Housing Market (Housing_v2.csv)	Price
10	F4640	Quebec Housing Sales price	sale_price
11	F8679	Medical Cost Personal Datasets	charges
12	F8683	Used Car Price Prediction	price_usd
13	F0479	Melbourne Housing Market	Price
14	E8979	Philippines Housing Market (Housing_v2.csv)	Price
15	F0104	Quebec Housing Sales price	sale_price
16	F8779	Medical Cost Personal Datasets	charges
17	F4639	Used Car Price Prediction	price_usd
18	F0686	Melbourne Housing Market	Price
19	F8723	Philippines Housing Market (Housing_v2.csv)	Price
20	F0438	Quebec Housing Sales price	sale_price
21	F8796	Medical Cost Personal Datasets	charges
22	F8761	Used Car Price Prediction	price_usd
23	F4890	Melbourne Housing Market	Price
24	F0133	Philippines Housing Market (Housing_v2.csv)	Price
25	E8948	Quebec Housing Sales price	sale_price
26	E9194	Medical Cost Personal Datasets	charges
27	E7961	Used Car Price Prediction	price_usd
28	F0484	Melbourne Housing Market	Price
29	F7320	Philippines Housing Market (Housing_v2.csv)	Price
30		Quebec Housing Sales price	sale_price
31		Medical Cost Personal Datasets	charges
32		Used Car Price Prediction	price_usd

Dataset	Link
Medical Cost Personal Datasets	https://www.kaggle.com/datasets/mirichoi0218/insurance
Used Car Price Prediction	https://www.kaggle.com/datasets/therohithanand/used-car-price-prediction
Melbourne Housing Market	https://www.kaggle.com/datasets/anthonypino/melbourne-housing-market
Philippines Housing Market (Housing_v2.csv)	https://www.kaggle.com/datasets/klekzee/phillipines-housing-market?select=Housing_v2.csv
Quebec Housing Sales price	https://www.kaggle.com/datasets/rickyjackson/quebec-housing-sales-price

2. **Data Quality Analysis**
    a. Provide a summary of the data set analysis:
        - For continuous variables, provide: total number of values, percentage of missing values, cardinality, minimum and maximum values, 1st and 3rd quartiles, mean, median, standard deviation.
        - For categorical variables, provide: total number of values, percentage of missing values, cardinality, mode, mode frequency value, mode percentage value, mode 2, mode 2 frequency value, mode 2 percentage value.
    b. Provide histograms of continuous variables and comment on what distribution it is.
    c. Identify data quality issues: missing values, outliers, and more. Provide ways for how these issues will be addressed and arguments to justify this choice.
    d. Identify relationships between variables using visualization tools such as scatter plot, SPLOM, bar plot, and more.
    e. Provide a correlation matrix for continuous variables and comment on the results.
    f. Prepare data for a linear regression model: perform normalization, convert categorical variables to continuous variables, add derived variables, and more. **For data transformation (normalization, modification of outliers, filling of missing values, etc.) the parameters obtained from the training sample must be used.*
3. **Linear Regression Model**
    a. Implement the calculation of the coefficients of the linear multivariable regression model using the following methods:
        i. By the method of least squares
        ii. Gradient descent method (or gradient descent method modifications)
    b. Create a linear regression model and calculate the coefficients using both realized methods. Present the coefficients for both models in a way that allows for easy comparison (e.g., in a table where positions correspond to the coefficients of the same variables). Compare the results.
    c. Test the hypotheses about the significance of the coefficients and modify the model if necessary.
    d. Cross-validate the model with at least 5 subsets of the data using the coefficient estimation method of your choice.
    e. Perform k-fold cross validation with at least 5 subsets for the cases:
        i. Only significant variables are included in the model
        ii. Only 3 most significant variables are included in the model
        iii. All possible variables are included in the model (if there were unsignificant ones)

For each case, calculate the mean and standard deviation of the selected error metrics and present them in a table. Visualize the results for each case using a box plot (displaying the plots for all cases on the same axes). Comment on the results: identify which model exhibits higher errors, which shows greater dispersion, and so on.

f. Evaluate the accuracy of models on test data - calculate error metrics, provide error analysis graphs, etc.
g. Implement the model using the standard functions of scikit-learn or another library. Compare the results of the models.
h. Prepare the selected model (with coefficients calculated using the least squares method or gradient descent) for use: it must be possible to input a single record in the original format, upon which data transformation and target variable prediction are performed.